

## Multiple Choice (10 marks)

1. In a commensalistic relationship between two species living symbiotically what are the consequences of their interactions?
  - (a) Both species benefit equally
  - (b) One species benefits, and the other is significantly harmed
  - (c) One species harms the other without any benefit
  - (d) Both species harm each other
  - (e) The species exchange benefits, but one benefits more than the other
2. Which of the following factors does NOT promote linkage disequilibrium?
  - (a) Asexual reproduction
  - (b) Coadapted gene complexes
  - (c) Epistasis
  - (d) Random mating
3. Which of the following is the correct characteristic of arteries?
  - (a) They carry blood towards the heart.
  - (b) They always carry deoxygenated blood.
  - (c) They carry blood away from the heart.
  - (d) They contain valves that prevent backflow.
  - (e) They always carry oxygenated blood.
4. What are some of the cues used by animal in migrating?
  - (a) Taste, wind direction, humidity
  - (b) Atmospheric oxygen levels, plant growth, tidal patterns
  - (c) Infrared radiation, sound frequency, barometric pressure
  - (d) Moon phases, echo location, gravitational pull
  - (e) Sunlight intensity, water current, soil composition
5. Adaptations of plants to minimize water loss in environments with high temperatures include which of the following?
  - (a) Thick, lush foliage
  - (b) Smooth, glossy leaves
  - (c) High rates of transpiration
  - (d) Shallow root systems
  - (e) Spines instead of leaves

## True / False (10 marks)

*Answer True or False*

6. True or False: Milk spoils exclusively due to the separation of fat content at lower temperatures, not due to bacterial activity.
7. True or False: An endotherm maintains a relatively stable metabolic rate despite variations in environmental temperature.
8. True or False: If  $46\%$  of the loci in a species' gene pool are heterozygous, the average homozygosity would be  $36\%$ .
9. True or False: The neritic zone is a marine ecosystem region where solar radiation penetrates and nutrients are often upwelled.
10. True or False: Muscle tissue is not classified as connective tissue; it is a distinct type of tissue responsible for movement.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss how genetic crosses involving strains of highly inbred mice can elucidate the genetic basis of transplant rejection or acceptance.
12. Discuss the fundamental nutritional requirements shared by all living organisms, and analyze the differences between phototrophs, chemotrophs, autotrophs, and heterotrophs.

*End of Paper*